

Question 01:

What does MATLAB stand for ?

- A. MathWorks
- B. Matrix Laboratory
- C. Math Laboratory
- D. None of the above

Answer: B

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Explanation:

The name MATLAB stands for **MATrix LABoratory**. MATLAB was written originally to provide easy access to matrix software developed by the LINPACK (linear system package) and EISPACK (Eigen system package) projects.

Question 02:

What symbol precedes all comments in MATLAB?

- A. “
- B. %
- C. //
- D. <
- E. None of the above

Answer: E

MATLAB is an interactive program for numerical computation and data visualization. You can enter a command by typing it at the MATLAB prompt '>>' on the Command Window.

...

Commands for Managing a Session

Command	Purpose
clc	Clears command window.
clear	Removes variables from memory.
exist	Checks for existence of file or variable.
global	Declares variables to be global.
help	Searches for a help topic.
lookfor	Searches help entries for a keyword.
quit	Stops MATLAB.
who	Lists current variables.
whos	Lists current variables (long display).

Commands for Working with the System

MATLAB provides various useful commands for working with the system, like saving the current work in the workspace as a file and loading the file later.

It also provides various commands for other system-related activities like, displaying date, listing files in the directory, displaying current directory, etc.

Command	Purpose
cd	Changes current directory.
date	Displays current date.
delete	Deletes a file.
diary	Switches on/off diary file recording.
dir	Lists all files in current directory
load	Loads workspace variables from a file.
path	Displays search path.
pwd	Displays current directory.
save	Saves workspace variables in a file.
type	Displays contents of a file.
what	Lists all MATLAB files in the current directory.
wklread	Reads .wkl spreadsheet file.

Input and Output Commands

MATLAB provides the following input and output related commands

Command	Purpose
disp	Displays contents of an array or string.
fscanf	Read formatted data from a file.
format	Controls screen-display format.
fprintf	Performs formatted writes to screen or file.
input	Displays prompts and waits for input.
;	Suppresses screen printing.

Format Code	Purpose
%s	Format as a string.
%d	Format as an integer.
%f	Format as a floating point value.
%e	Format as a floating point value in scientific notation.
%g	Format in the most compact form: %f or %e.
\n	Insert a new line in the output string.
\t	Insert a tab in the output string.

Question 02:

What symbol precedes all comments in MATLAB?

- A. “
- B. %
- C. //
- D. <
- E. None of the above

Answer: E

Explanation:

MATLAB is an interactive program for numerical computation and data visualization. You can enter a command by typing it at the MATLAB prompt '>>' on the Command Window.

...

Question 03:

Which of the following is not a pre-defined variable in MATLAB.

- A. Pi
- B. Inf
- C. I
- D. Gravity
- E. j

Answer: D

Question 04:

Which MATLAB command clears all data and variables stored in memory?

- A. Clc
- B. Clear
- C. Delete
- D. Deallocate
- E. None of the above

Answer: B

Question 04:

Which MATLAB command clears all data and variables stored in memory?

- A. Clc
- B. Clear
- C. Delete
- D. Deallocate
- E. None of the above

Answer: B

Explanation:

To clear all variables from the current workspace, use **clear** or **clearvars** . To clear all global variables, use **clear global** or **clearvars -global** .

clear FUNCTIONS removes all compiled MATLAB and MEX-functions. **clear ALL** removes all variables, globals, functions and MEX links. **clear ALL** at the command prompt also clears the base import list.

Question 05:

MATLAB was developed by

- A. Albert Einstein
- B. Cleve Moler
- C. Marie Curie
- D. Isaac Newton

Answer: B

Explanation:

MATLAB was invented by **mathematician and computer programmer Cleve Moler**. The idea for MATLAB was based on his 1960s PhD thesis. Moler became a math professor at the University of New Mexico and started developing MATLAB for his students as a hobby.

Question 06:

..... Is used to check if two elements are equal in MATLAB?

- A. !=
- B. ==
- C. Isequal
- D. =

Answer: B

Explanation:

It's **used to compare two variables** (numbers, arrays, etc.) and say whether they match or not. This is different than a single = sign, which is an assignment. In `a==b` you'll get a single true or false value, or an array of them if `a` and `b` are arrays.

`A == B` returns a logical array with elements set to logical 1 (true) where arrays `A` and `B` are equal; otherwise, the element is logical 0 (false). The test compares both real and imaginary parts of numeric arrays.

Question 07:

Keys combination used to stop execution of a command in MATLAB?

- A. ctrl+c
- B. ctrl+s
- C. ctrl+b
- D. ctrl+enter

Answer: A

Question 07:

Keys combination used to stop execution of a command in MATLAB?

- A. ctrl+c
- B. ctrl+s
- C. ctrl+b
- D. ctrl+enter

Answer: A

Explanation:

To stop execution of a MATLAB® command, press **Ctrl+C** or **Ctrl+Break**.

Question 08:

Which class is used in MATLAB to store the complex number?

- A. Double
- B. Symbolic
- C. Character
- D. Array

Answer: B

Explanation:

The complex number represented in MATLAB uses 'i' as a symbolic character, the class of any complex number is **symbolic**.

Question 09:

Which of the following is a Assignment operator in MATLAB?

- A. '+'
- B. '='
- C. '*'
- D. '\$'

Answer: B

Explanation:

Question 10:

Which of the following is a switch-case structure in MATLAB?

- A. Logical Structure
- B. Conditional Structure
- C. Hierarchical Structure
- D. Multidimensional Structure

Answer: A

Question 11:

What are the features of MATLAB ?

1. Full graphics capabilities
2. Platform Independent
3. Ease of Use & Good Source of Help
4. All of the above

Answer: All of the above

Explanation:

MATLAB (matrix laboratory) is a **fourth-generation high-level programming language and interactive environment for numerical computation, visualization and programming.** ...

It has numerous built-in commands and math functions that help you in mathematical calculations, generating plots, and performing numerical methods.

Features of MATLAB

1. **MATLAB is a high-level language:** MATLAB Supports Object oriented programming.
2. **Interactive graphics:** MATLAB has inbuilt graphics to enhance user experience.
3. **A large library of Mathematical functions:** MATLAB has a huge inbuilt library of functions required for mathematical analysis of any data. It has common math functions like sqrt, factorial etc.
4. **Data access and processing:** MATLAB allows accessing of data from external sources like image files (.jpg, .PNG), audio files (.mp), and real-time data from JDBC/ ODBC.
5. **Interactive environment:** MATLAB offers interactive environment by providing a GUI (Graphical user interface) and different types of tools like signal analyses and tuners.
6. **MATLAB can interface with different languages:** We can write a set of codes (libraries) in languages like PERL and JAVA, and we can call those libraries from within the MATLAB itself.
7. Being compiler independent makes MATLAB more efficient and productive

Question 12:

Which of the following is the disadvantage of MATLAB?

1. MATLAB is a fourth-generation high-level language
2. Being compiler independent makes MATLAB more efficient and productive
3. MATLAB codes are written in sentences and executed one by one
4. MATLAB is an interpreted language thus it can be very slow

Answer:

MATLAB is an interpreted language thus it can be very slow.

Disadvantages of MATLAB

- MATLAB is slow since it is an interpreted language that is MATLAB programs are not converted into Machine language but are run by external software, so it can sometimes be slow.
- We cannot create the OUTPUT file in MATLAB.
- One cannot use graphics in MATLAB with -nojvm option, on doing so, we will get a runtime error.
- We cannot make functions in one single .m file as we have in the case of other programming languages. We have to create different files for different functions.
- Sometimes, the error messages are not much informative, so you have to figure out the error yourself.

Question 13:

What is an M file in MATLAB?

- A. Extension file
- B. Design file
- C. Simple text file.
- D. Function file

Answer: C

Explanation:

What is an M file in MATLAB?

An m-file, or script file, is **a simple text file where you can place MATLAB commands**. When the file is run, MATLAB reads the commands and executes them exactly as it would if you had typed each command sequentially at the MATLAB prompt. All m-file names must end with the extension '. m'

The M Files

MATLAB allows writing two kinds of program files –

- ▣ **Scripts** – script files are program files with **.m extension**. In these files, you write series of commands, which you want to execute together. Scripts do not accept inputs and do not return any outputs. They operate on data in the workspace.
- ▣ **Functions** – functions files are also program files with **.m extension**. Functions can accept inputs and return outputs. Internal variables are local to the function.

You can use the MATLAB editor or any other text editor to create your **.mfiles**. In this section, we will discuss the script files. A script file contains multiple sequential lines of MATLAB commands and function calls. You can run a script by typing its name at the command line.

Question 14:

To add a comment to the mfile, the MATLAB command is

1. %
2. ;
3. Comment ('')
4. &

Answer: %

Explanation:

To add comments to MATLAB code, use the **percent (%) symbol**. Comment lines can appear anywhere in a code file, and you can append comments to the end of a line of code.

Question 15:

What is use of ABS function in MATLAB ?

1. to return absolute value
2. to run absolute value
3. to retain absolute value
4. to revert absolute value

Answer: to return absolute value

Explanation:

abs() function is used **to return absolute value in MATLAB**. It can be represented as: $Y = \text{abs}(X)$.

the **absolute value** or modulus of a real number x , denoted $|x|$, is the non-negative value of x without regard to its sign.

Question 16:

To join one or more string into a single string is known as

1. Concatenation
2. Joining
3. String conversion
4. String theory

Answer: Concatenation

Explanation:

To join the strings together you just need to construct a **string concatenation**: `allStrings.concat(string1, string2, ..., stringX)`

Question 17:

To print a newline in a fprintf statement, you must use the following escape character:

1. \t
2. \nl
3. \nxt
4. \n

Answer: \n

fprintf means in MATLAB

The fprintf function allows you to "write" information to the screen for the user to view. This very important when user interaction is involved. The 'f' in printf stands for formatted. This means you can "format" how the data is printed in such a manner as to make it easy to read.

\n inserts a newline character.

What is the difference between printf and fprintf in Matlab?

printf is a C function to print a formatted string to the standard output stream which is the computer screen. fprintf is a C function to print a formatted string to **a file**. 03-Jan-2018

Question 18:

How do you enter a comment in a program?

1. //.....\\
2. /'.....'\
3. /*.....*\
4. /.....\

Answer: /*.....*\

Explanation:

A comment **starts with a slash asterisk /* and ends with a asterisk slash */** and can be anywhere in your program. Comments can span several lines within your C program. Comments are typically added directly above the related C source code.

Question 19:

The num2str command

1. Convert a number to string
2. Convert a string to number
3. Concatenates numbers and string
4. Concatenates string

Answer: Convert a number to string

Explanation:

The num2str function **converts numbers to their string representations**. This function is useful for labeling and titling plots with numeric values. `str = num2str(a)` converts array `A` into a string representation `str` with roughly four digits of precision and an exponent if required.

Question 20:

Which of the following is the output of `Cat=['cat' 'dog']` is

1. Catdog
2. Cat dog
3. Cat&dog
4. CatDog

Answer: Catdog

Question 21:

Index of an array in MATLAB start with

1. 0
2. 1
3. Depends on the class of array
4. unknow

Answer: 1

Explanation:

MATLAB indices start from **1 (linear indexing)**, which is standard in mathematics (and matrix manipulation in particular).

In most programming languages, the first element of an array is element 0. In MATLAB, **indexes start at 1**.

Question 22:

What is the value of ans that is printed when the following code is run:

`isnumeric(32)`

- 1. 1
- 2. 0
- 3. 32
- 4. True
- 5. yes

Answer: 1

Explanation:

`isnumeric` is a predefined function that determines whether the input is a numeric array or not.

`isnumeric(A)` returns logical 1 (true) if A is an array of numeric data type. Otherwise, it returns logical 0 (false).

Numeric types in MATLAB® include: `int8`, `int16`, `int32`, `int64`, `uint8`, `uint16`, `uint32`, `uint64`, `single`, and `double`.

Question 23:

When used in the fprintf command, the %g is used as the

1. Single character display
2. Fixed point display
3. String notation display
4. Default number display

Answer: Default number display

Available Conversion Characters As %f For fprintf() Command In MATLAB

- %f**: This takes all decimal values inside sentences. You can define the precision of decimal values by adding precision numbers which are shown below.
- %e** or **%E**: Exponential notation which is also called scientific notation that you can use (For example 3.3526e+000).
- %g**: If you use this conversion character, the trailing zeros for decimal numbers will not be shown.
- %u**, **%d**, and **%i**: This conversion character shows the value of the variable number in base 10.
- %o**: This conversion character shows the value of the variable number in base 8 which is also the first letter of octal.
- %x**: This conversion character shows the value of the variable number in base 16 in which the numbers between 10-16 are shown letters between a – f.
- %c**: Shows single characters which is the value of a variable in Matlab®. This can be useful when you take string information from users.
- %s**: This is an actual string array that is different from %c.

Question 24:

In MATLAB, this keyword immediately moves to the next iteration of loop:

1. Update
2. goto
3. Continue
4. break

Answer: Continue

Explanation:

continue passes control to the next iteration of a for or while loop. It skips any remaining statements in the body of the loop for the current iteration. The program continues execution from the next iteration.

Question 25:

Characters in MATLAB are represented in their value in memory

1. Decimal
2. ASCII
3. Hex
4. String

Answer: ASCII

Explanation:

ASCII is a **7-bit character set containing 128 characters**. It contains the numbers from 0-9, the upper and lower case English letters from A to Z, and some special characters. The character sets used in modern computers, in HTML, and on the Internet, are all based on ASCII.

Computer can only understand binary no. , so an ASCII code is the internal representation of a characters. MATLAB has two useful function to work with them: Characters and double

Question 26:

Which of the following will correctly define X,Y and Z as a symbols?

1. `sym (x, y, z)`
2. `sym (x y z)`
3. `syms x, y, z`
4. `sym x, y, z`

Answer: `sym (x y z)`

Question 27:

If I want to save a formatted string to memory, but don't want to print it out, which command should I use?

1. fprintf
2. sprintf
3. Disp
4. echo

Answer: sprintf

Explanation:

Use **sprintf**. `int sprintf ( char * str, const char * format, ... );`
Write formatted data to string Composes a string with the same text that would be printed if format was used on printf, but instead of being printed, the content is stored as a C string in the buffer pointed by str.

Question 28:

Which of the following commands is used to determine whether an input in MATLAB is a keyword?

1. iskeyword function
2. inkeyword function
3. waskeyword function
4. thekeyword function

Answer: iskeyword function

Explanation:

To determine if the input is a MATLAB keyword, use **the iskeyword function**. `isvarname` is the command form of the syntax. The command form requires fewer special characters. You do not need parentheses or single quotes around the input.

Question 29:

To better manage memory and prevent unnecessary memory allocations, MATLAB uses:

1. Vectors
2. Scalars
3. Matrix math
4. Delayed Copy

Answer: Delayed Copy

Question 30:

Which MATLAB command clears all data and variables stored in memory?

- A. Clc
- B. Clear
- C. Delete
- D. Deallocate
- E. None of the above

Answer: B) Clear

Explanation:

To clear all variables from the current workspace, use **clear** or **clearvars** . To clear all global variables, use **clear global** or **clearvars -global** .

Clear FUNCTIONS removes all compiled MATLAB and **MEX-functions**. **clear ALL** removes all variables, globals, functions and MEX links. **clear ALL** at the command prompt also clears the base import list.

MATLAB

- Is a high-performance language
- Stands for *Matrix Laboratory*
- **MATLAB** was developed by mathematician and computer programmer *Cleve Moler*.
- It Can be used as a fancy calculator
- Allows you to easily work with entire matrices rather than one number at a time
- Is useful for anything that requires matrix and vector manipulations such as:
 - Mathematical, scientific, engineering, statistical and financial problems
 - Anything that requires plotting/visualizing and analyzing data
- Comes with a basic set of tools for visualizing data and for performing calculations on matrices and vectors
 - For specific technologies, MATLAB provides toolboxes, which add to the basic MATLAB functionality. We have the **Image Processing Toolbox**. Some other toolboxes include:
 - Statistics Toolbox
 - Neural Network Toolbox
 - Fuzzy Logic Toolbox
 - Signal Processing Toolbox
 - Wavelet Toolbox
 - Financial Toolbox
 - Bioinformatics Toolbox
 - Database Toolbox

The MATLAB screen mainly include 4 parts:

Command Window, Workspace window, Command History window and Current Directory window.

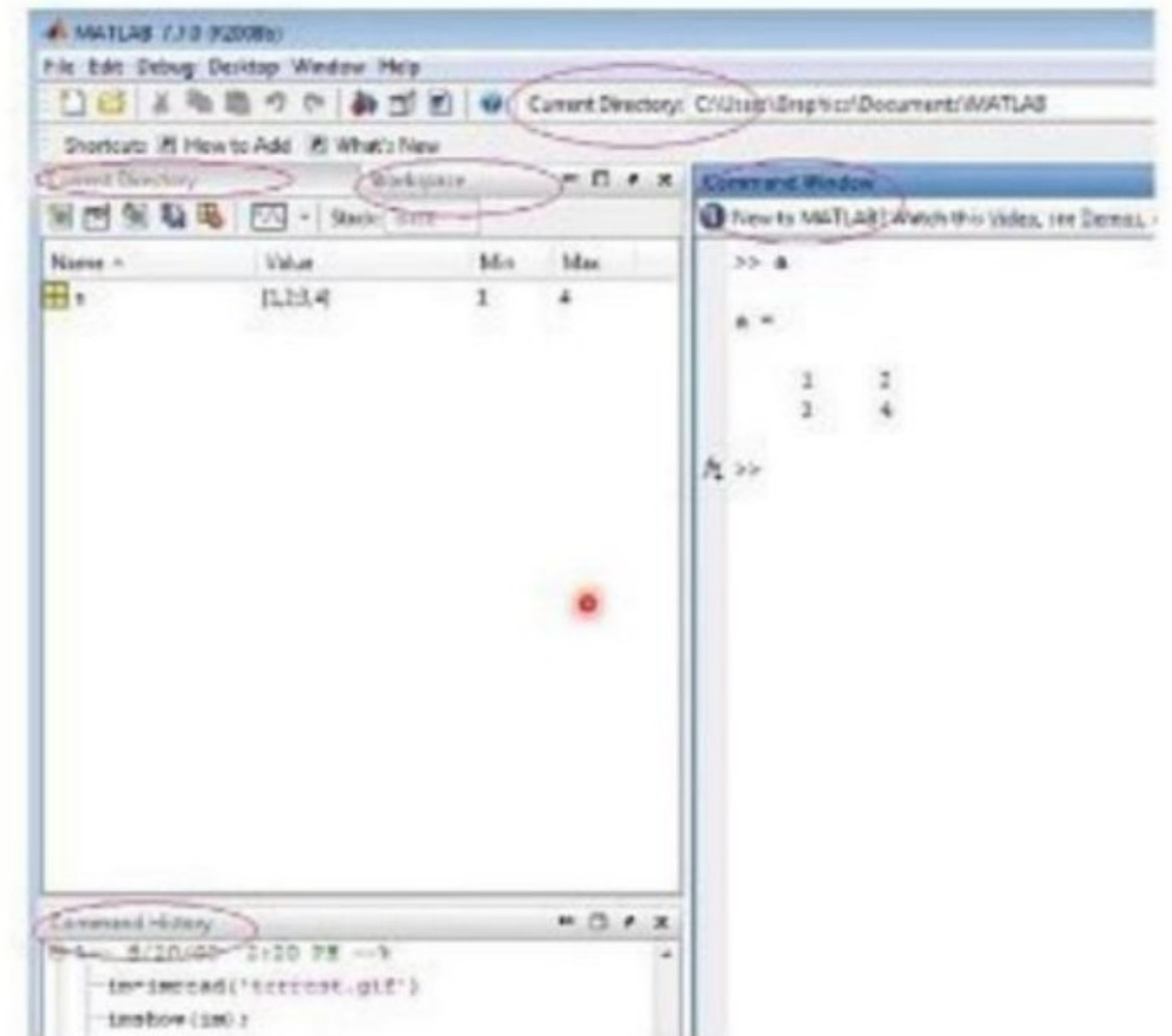
Command Window – this is where you type commands and usually the outputs (or error messages) appear here too.

Workspace window – if you define new variables there names should be listed here.

Command History window – this is where past commands are remembered. If you want to re-run a previous command or to edit it you can drag it from this window to the command window or double click to re-run it.

Current Directory window- show the files in Current Directory.

Launch Pad Window- to access all MATLAB services and toolboxes.



Question 01:

The latest version of MATLAB is

1. MATLAB R2020a.
2. MATLAB R2020b.
3. MATLAB R2021a.
4. MATLAB R2021b.

Answer: MATLAB R2021a.

Explanation:

Latest version of MATLAB is **MATLAB R2021a.**

Question 02:

Which is the invalid variable name in MATLAB?

1. x6
2. last
3. 6x
4. z

Answer: 6x

Explanation:

In MATLAB variable is not start with number
eg. 6x is invalid

Question 03:

When used in the `fprintf` command, the `\n` is used to

1. Add a space between any two characters
2. Start a line space/line (enter key)
3. Place the number into the comment
4. Clear the comment

Answer: start a line space/line (enter key)

Explanation:

The `fprintf` function is used for printing information to the screen. The `fprintf` function prints an array of characters to the screen:

```
fprintf("Happy Birthday\n"); ...
```

fprintf command: The fprintf function is used for printing information to the screen. The fprintf function prints an number of characters to the screen:

```
fprintf('Happy Birthday\n');
```

When using the % signs to print out the data stored in variables, we must use the same number of % signs as the number of **variables**.

Some common format options:

- %s - print a string
- %c - print a single character
- %d - print a whole number
- %f - print a floating point number
- %g – default number display
- \n - print a new line (go to the next line to continue printing)
- \t - print a tab
- \\ - print a slash
- %% - print a percent sign

Question 04:

To display 'Question 2' in the command window, the correct command is

1. `disp(Question 2)`
2. `display('Question 2')`
3. `disp('Question 2')`
4. `Question 2`

Answer: `disp('Question 2')`

Question 05:

When used in the fprintf command, the %g is used as the ?

1. Single character display
2. Fixed point display
3. String notation display
4. Default number display

Answer: Default number display

Explanation:

For the %g operator, the precision indicates the number of significant digits to display. For the %f, %e, and %E operators, the precision indicates how many digits to display to the right of the decimal point. Display numbers to different precisions using the precision field.

Question 06:

Which of the following is the program file in MATLAB?

1. Scripts
2. Text
3. Function
4. Both 1 & 3

Answer: Both 1 & 3

Explanation:

Scripts file: It is a simple text file where you can place **MATLAB commands**. When the file is run, MATLAB reads the commands and executes them exactly as it would if you had typed each command sequentially at the MATLAB prompt. All m-file names must end with the extension '.m'

Function file: MATLAB® program files can contain code for more than one function. In a function file, the first function in the file is called **the main function**. This function is visible to functions in other files, or you can call it from the command line.

Question 07: Which of the following function is not performed by Simulink?

1. System level design
2. Simulation
3. Automatic code generation
4. None of these

Answer: None of these

Explanation:

Use MATLAB and Simulink together to combine the power of textual and graphical programming in one environment.

Simulink is graphical and more interactive to the user. Whereas the Matlab is coding based approach based on the different function available in Matlab.

Question 08:

Which of the following window of MATLAB is used to create and debugs scripts and function files.

1. Command Window
2. Figure Window
3. Editor Window
4. Launch Pad Window

Answer: Editor Window

Explanation:

The MATLAB Editor Window is a **simple text editor where you can load, edit and save complete MATLAB programs**. The Editor window also has a menu command (Debug/Run) which allows you to submit the program to the command window.

Question 09:

The terms “LINPACK” and “EISPACK associated with MATLAB stands for”

1. Location system package & Europe system package
2. Location system package & Eigen system package
3. Linear system package & Emulator system package
4. Linear equation package & Eigen system package

Answer: Linear equation package & Eigen system package

Explanation:

LINPACK and EISPACK are Fortran subroutine packages for matrix computation developed in the 1970's.

Question 10:

Which of the following is used for Logical OR operation in MATLAB?

1. &
2. |
3. ==
4. ~

Answer: |

Explanation:

IN MATLAB

	Matlab Operator	Notes
Arithmetic Operator	+	Additions
	-	Reductions
	/	Divisions
	*	Multiplication
	^	Rank
Relational Operator	==	Equal to
	~=	Not equal to
	<	Less than
	<=	Less than or equal to
	>	More than
	>=	More than or equal to
Logical Operator	&	And
		Or

Question 11:

How do you convert a number to a string in MATLAB?

1. Str function
2. Ptr function
3. Ltr function
4. Ctr function

Answer: Str function

Explanation:

Convert to Strings

Before R2016b, convert to character vectors using `num2str`.

To convert a number to a string that represents it, use the `string` function.

```
str = string(pi)
```

```
str =  
"3.1416"
```

The `string` function converts a numeric array to a string array having the same size.

```
A = [256 pi 8.9e-3];  
str = string(A)
```

```
str = 1x3 string  
    "256"    "3.141593"    "0.0089"
```

You can specify the format of the output text using the `compose` function, which accepts format specifiers for precision, field width, and exponential notation.

```
str = compose("%9.7f",pi)
```

```
str =  
"3.1415927"
```

Question 12: Which command is used in MATLAB workspace to display listing ?

1. dir
2. cd
3. clc
4. type

Answer: dir

Explanation:

Command	Purpose
cd	Changes current directory.
date	Displays current date.
delete	Deletes a file.
diary	Switches on/off diary file recording.
dir	Lists all files in current directory.
load	Loads workspace variables from a file.
path	Displays search path.
pwd	Displays current directory.
save	Saves workspace variables in a file.
type	Displays contents of a file.
what	Lists all MATLAB files in the current directory.
wklread	Reads .wk1 spreadsheet file.

Command	Purpose
clc	Clears command window.
clear	Removes variables from memory.
exist	Checks for existence of file or variable.
global	Declares variables to be global.
help	Searches for a help topic.
lookfor	Searches help entries for a keyword.
quit	Stops MATLAB.
who	Lists current variables.
whos	Lists current variables (long display).
Command	Purpose
disp	Displays contents of an array or string.
fscanf	Read formatted data from a file.
format	Controls screen-display format.
fprintf	Performs formatted writes to screen or file.
input	Displays prompts and waits for input.
;	Suppresses screen printing.

Question 13:

Mention the function available in MATLAB to design the PID (proportional-integral-derivative) controller.

1. Pid (plant name)
2. Tuner (plant name)
3. pidTuner (plant name)
4. Step (T,t)

Answer: pidTuner (plant name)

Explanation:

C = pid(Kp , Ki , Kd) creates a proportional, integral, and derivative (PID) controller with $T_f = 0$. **C = pid(..., Name, Value)** creates a controller or converts a dynamic system to a pid controller object with additional options specified by one or more Name, Value pair arguments.

Question 14:

Identify the **WRONG** statement.

1. MATLAB has huge built-in library of functions.
2. MATLAB is free and hence users need not obtain licensed version.
3. MATLAB provides tool to develop GUI based applications.
4. MATLAB is platform independent.

Answer:

MATLAB is free and hence users need not obtain licensed version

Question 15:

Simulink is the GUI based companion software for.....

1. ETAP
2. LabView
3. MAGNET
4. MATLAB

